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# Information Disclosure and Competitive Dynamics: Evidence from the Pharmaceutical Industry

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**Abstract.** This paper studies how competitive dynamics shape innovative firms' voluntary disclosure of product quality information. Our empirical context is the pharmaceutical industry, where firms must decide whether to disclose private drug quality information acquired in clinical trials. Using a difference-in-difference strategy, we show that the approval of a competitor's drug lowers the likelihood of a firm reporting its clinical trial results by 13%. We explore how these effects vary based on the project quality, competitor type, and firm experience. These findings suggest that strategic considerations play a role in firms' disclosure decisions; in response to a competitor's drug approval, firms may selectively withhold information to maintain and improve their competitive position.

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## 1. Introduction

Firms in a wide range of innovation-driven industries, such as automobile manufacturing, software, and healthcare, generate private information about the quality of their products. For instance, automobile manufacturers conduct crash tests to generate vehicle safety information; developers conduct software tests to assess product performance; and pharmaceutical firms invest in research and development (R&D) projects, such as clinical trials, to generate information about the safety and efficacy of their drugs. In leading theoretical models, firms have incentives to voluntarily disclose all private product quality information (Grossman 1981, Milgrom 1981, Jovanovic 1982).

However, the complete disclosure of private information is rarely observed in practice (see Milgrom 2008 and Dranove and Jin 2010 for surveys). For example, information generated from clinical trials is of substantive interest as it is used to shape the decisions of regulators, consumers, and competitors. Yet, a large policy-oriented literature has conjectured that firms engage in strategic disclosure by withholding or sharing clinical trial results from key stakeholders (e.g., consumers making product adoption decisions, rivals developing similar products) to maintain their competitive advantage (see, e.g., Lexchin et al. 2008 and Anderson et al. 2015). Motivated by concerns about transparency, Congress passed regulations in 2007 requiring detailed public results reporting on the federal registry, ClinicalTrials.gov, within one year

of trial completion (U.S. Food and Drug Administration 2007).<sup>1</sup> Nevertheless, a 2011 analysis found that two years after 5,600 clinical trials were completed, just 22% of them had reported their results on the site (Prayle et al. 2012).

Despite a growing number of policies aimed at motivating firms to disclose their trial results, widespread underreporting persists, with “competitive pressures” often cited as key drivers (Berenson 2005, Zarin et al. 2011). Although existing empirical analyses have separately shed light on the causes and consequences of firms' disclosure practices and competitive dynamics (see, e.g., Dranove and Jin 2010 and Cookson 2018), very little is known as yet about how firms' disclosure decisions are shaped by changes in their competitive landscape. To fill this gap, we leverage the unique features of the pharmaceutical R&D setting to examine whether and how firms adjust their information provision in response to a change in their market environment. In particular, we utilize the comprehensive documentation of R&D activity and the staggered arrival of competitor drug approvals to identify the causal relationship between competitor drug approvals and firms' decision to disclose detailed clinical trial results. We focus on how competitor drug approvals shape two dimensions of disclosure: (i) the probability and (ii) the timing. We find that, on average, firms facing competitor drug approvals are less likely to disclose key clinical trial results, and if they do disclose, they are more likely to delay disclosure.

Much of the existing theoretical literature is mixed on how competitive dynamics might shape firms' disclosure incentives (Board 2009; Guo and Zhao 2009; Levin et al. 2009; Jansen 2010, 2017; Gentzkow and Kamenica 2017; Markopoulos and Hosanagar 2018). For example, if shifts in firms' competitive landscapes raise firm concerns about dampening consumer interest, they may withhold information to prevent consumers from making unfavorable comparisons. Conversely, in some cases, firms may choose to share information to signal their similarities to successful rivals. Closely related is the limited empirical literature on competition and product quality disclosure, which primarily relies on cross-sectional evidence. For example, Jin (2005) shows that markets with more health maintenance organizations have lower levels of product quality disclosure among such organizations. However, concerns about real-world and unobservable factors (e.g., market demand, technological opportunities) may limit the external validity and causal interpretation of existing estimates. One reason for the sparse empirical literature is that causal relationships are difficult to test as it requires observing the entire competitive environment and tracking products with disclosed and undisclosed information, and exogenous changes in firms' competitive landscape and their competitive expectations are rare.

Three features of the pharmaceutical industry allow us to overcome these constraints. First, diseases (e.g., ovarian cancer) provide a natural and distinct categorization of markets. Second, pharmaceutical R&D is subject to substantial regulatory oversight and competitive monitoring, which provides researchers with the data to observe each drug as it moves through the development cycle. This careful documentation allows researchers to observe a near "universe" of drugs in a disease market and to determine the exact date that detailed clinical trial results are publicly disclosed. Third, uncertainties in the regulatory approval process allow us to take advantage of variation in the plausibly exogenous timing of competitor drug approvals to identify the causal effect of such approvals on firms' disclosure incentives.

We assemble a rich data set of clinical trials initiated between 2007 and 2019, which we then link to data on firm and market characteristics.<sup>2</sup> Using these data, we conduct three types of analyses. First, we establish facts about disclosure practices in the pharmaceutical industry. A key challenge in investigating these practices is that we only have access to disclosed trial results, making it difficult to investigate how disclosure might relate to characteristics of R&D projects, such as quality. We address this challenge in two ways. First, we construct a proxy for project quality that is based on project success or failure.<sup>3</sup> In particular, we examine whether the trial's drug advances to the next development phase following trial completion (successful project) or does not advance (failed project). Monitoring which drug projects a firm

advances and which it deems a failure and terminates allows us to approximate the quality of all trials, regardless of whether their results are reported. Second, we systematically analyze reported clinical trial outcomes to identify patterns of selective disclosure, revealing that firms delay the release of less favorable results.

Second, we establish the causal effect of competitor drug approvals on firms' disclosure decisions. We employ a difference-in-difference (DiD) approach that leverages the timing at which the FDA approves promising competitor drugs: competitor drugs with "priority review status." The motivation behind this empirical analysis is that given that the exact timing of approval is subject to uncertainty, the approval of a rival priority review drug in a disease signals a significant (and plausibly exogenous) change in the competitive landscape, shifting competitive expectations of other firms conducting trials in that disease market (Gilchrist 2016). Our primary results indicate that a competitor's priority review drug approval lowers the likelihood of a firm reporting its trial results on ClinicalTrials.gov by 13%.

Third, we conduct heterogeneity analyses that shed light on the channels through which competitor drug approvals may shape firms' disclosure decisions. First, we show that competitor drug approvals lead to reduced disclosure for both failed and successful R&D projects, with the effects among failed projects greater than (although not statistically different from) the effects among successful projects. This is consistent with the view that competitor drug approvals lead firms to shift their competitive expectations, increasing their concerns about dampening consumer interest and the leakage of valuable information to competitors that can quickly leverage it. Second, we leverage technological differences across competitor drug approvals and document evidence suggesting that firms, when faced with a successful rival, strategically withhold bad news and disclose good news. Third, we document that smaller, less experienced firms, which are likely to be more impacted by shifts in the market environment, show larger shifts in their disclosure decisions. Finally, we provide evidence that suggests that competitor drug approval-driven changes in R&D investments are unlikely to be the primary channel through which such approvals shape disclosure. Overall, these results suggest that competitor approvals increase the strategic relevance of firms' trial results.

Our empirical focus on the pharmaceutical industry, an industry characterized by high levels of R&D activity, is of substantive interest as firm secrecy has been widely documented and timely access to information about the quality of novel products has significant consequences for drug manufacturers and their competitors, consumers, and regulators (Arrow 1963, Harris 2010). For competitors, having access to rivals' clinical trial outcomes can minimize R&D investments that are redundant or unlikely to lead to success (Zarin and Tse

2008). For physicians and patients, the publication of clinical trial results may reveal more effective treatments (Law et al. 2011). For policymakers aiming to spur innovation and reduce excessive spending, published results facilitate the efficient exchange of ideas (Teece 1986, Gans and Stern 2003, Arora et al. 2004) and reduce wasteful spending on ineffective products, which can cost up to \$28.6 billion annually (Eichler et al. 2013, Shrank et al. 2019).

Our analysis provides important insights for scholars and practitioners of business strategy (Guo and Zhao 2009). Our results clarify that strategic disclosure may be an important lever by which firms maintain and improve their relative performance. This finding adds to a growing literature that describes how in response to competitor drug approvals, firms shift decisions related to pricing (Goolsbee and Syverson 2008; Seamans 2012, 2013), product quality (Prince and Simon 2015), alliances (Goetz and Shapiro 2012), capacity (Lieberman 1987, Cookson 2018), and advertising (Morton 2000, Ellison and Ellison 2011). These results also extend the traditional literature on product quality information, which often centered on a single firm and whose findings were often independent of other key features of the environment, such as competition and firm experience (see, e.g., Milgrom 2008 and Dranove and Jin 2010).

The paper proceeds as follows. Section 2 describes the setting and the conceptual framework. Section 3 introduces the data. Section 4 shows descriptive facts, and Section 5 presents the paper's main results regarding the causal impact of competitor approvals on disclosure. Section 6 explores the mechanisms and provides additional heterogeneity analyses. Section 7 concludes.

## 2. Setting and Conceptual Framework

This section describes how the setting—clinical trial results reporting in the U.S. pharmaceutical industry—can speak to the relationship between competitive dynamics and strategic disclosure. Further, we motivate the empirical analysis by discussing the central prediction that results from the literature and the empirical setting.

### 2.1. Disclosure in the Pharmaceutical Industry

**2.1.1. Clinical Trials.** Accounting for nearly 17% of industrial R&D and 62% of industrial R&D spending with health applications (National Center for Science and Engineering Statistics and Census Bureau 2018), the U.S. pharmaceutical industry provides a useful setting to monitor firm disclosure of product quality information, particularly in the context of reporting clinical trial results. Characterized by high uncertainty and large capital investments, the drug development process typically begins with extensive preclinical laboratory

research that involves testing a new drug candidate on animals and human cells. When these tests demonstrate potential safety and efficacy, the manufacturer begins the increasingly expensive aspect of the drug's development: human testing of drugs in a series of clinical trials. Drugs that successfully demonstrate safety in small phase I clinical trials proceed to phase II clinical trials, which typically test their safety and efficacy in 80–200 participants. Successful drugs move on to phase III trials, which assess their safety and efficacy in several hundred or thousand participants over a substantial period, typically one to four years.

If their phase III clinical trials are successful, drug manufacturers interested in seeking regulatory approval of a novel drug to treat a specific disease (indication) must submit a New Drug Application (NDA) containing clinical trial data to the FDA. If the FDA approves the drug, it then releases summary reports of the drug's clinical findings. After its initial approval, a drug's manufacturer may further test its drug in additional phase II and phase III clinical trials.<sup>4</sup> For example, to increase demand, a firm with an approved drug for colorectal cancer may conduct additional trials to garner further efficacy results, seek approval of the drug in an additional ("supplemental") indication (e.g., in ovarian cancer), or explore off-label indications. Indeed, approximately 90% of FDA-approved drugs have additional therapeutic uses (Gelijns et al. 1998), and an estimated 20%–80% of drug use is for "off-label" (nonapproved) indications (Wittich et al. 2012). Although firms are not permitted to directly advertise off-label indications, if a firm is interested in expanding off-label use of the drug, it may use evidence generated from these clinical trials to indirectly promote additional indications to physicians (Kessler et al. 1994, Azoulay 2002, Shapiro 2018).

Importantly, regardless of their purpose (e.g., to support an initial approval or a supplemental approval) or outcome (e.g., failure and the termination of a drug project), clinical trials are regulated by the FDA. Under FDA regulations, the sponsors of any nonphase I trials register these trials on the National Institutes of Health (NIH) website ClinicalTrials.gov, a public registry, and report their safety and efficacy results when they are completed (Department of Health and Human Services 2016).<sup>5</sup>

**2.1.2. ClinicalTrials.gov.** Established under the Food and Drug Administration Modernization Act of 1997, ClinicalTrials.gov is considered the world's largest clinical trial registry, and it is seen as providing a near census of nonphase I clinical trials (Zarin et al. 2011). Under existing ClinicalTrials.gov requirements, when its trial is first initiated, the drug manufacturer (i.e., the trial sponsor) must prospectively report basic clinical trial information (e.g., its purpose, design, patient eligibility criteria, and location).<sup>6</sup> To further increase



transparency between trial sponsors and key stakeholders, in 2007, Congress passed the Food and Drug Administration Amendments Act (FDAAA), requiring sponsors to submit key trial outcomes (e.g., primary end points) in a series of simple forms that ClinicalTrials.gov then publicly posts in a standardized and structured format (see Figure A1 in the Online Appendix).<sup>7</sup> With some exceptions, clinical trial sponsors must submit their trial findings no later than one year after the date of final data collection for the primary outcome measure (the “primary completion date”).<sup>8</sup> Consequently, since 2007, ClinicalTrials.gov has provided a database of critical trial efficacy and safety outcomes for nonphase I trials (Zarin et al. 2016). This database is the world’s largest clinical trial results repository, and it has been shown to have meaningful effects in shaping subsequent firm decisions (Aghamolla and Thakor 2022, Hsu et al. 2024). However, although the *registration* of clinical trials has become standard practice, concerns about the selective *reporting* of their results persist (Bourgeois et al. 2010, Hudson et al. 2016, Kao et al. 2023).

**2.1.3. Regulatory Concerns.** Concerned that firms may strategically withhold trial results, regulators have enacted several additional efforts to incentivize reporting on ClinicalTrials.gov. For example, failure to submit clinical trial results on ClinicalTrials.gov can lead to FDA-issued fines of \$10,000 a day or withholding of NIH grant funds (U.S. Food and Drug Administration 2007). In addition, firms that withhold clinical findings may trigger costly litigation (Marinovic and Varas 2016). For example, in February 2021, Bristol Myers Squibb and Sanofi were ordered to pay \$834 million to the State of Hawaii for violating consumer protection laws by failing to disclose that their drug, Plavix, was less effective for people of Asian or Pacific Island descent in clinical trials.<sup>9</sup>

Despite the threat of a financial penalty, results reporting on ClinicalTrials.gov remains low. For example, Anderson et al. (2015) examined 13,000 clinical trials conducted prior to September 2013 and found that only 38% of them had reported results. In addition to strategic considerations, such as competitors’ actions, another reason for the relatively low levels of clinical trial results reporting may include the perceived low risk of penalties for nonreporting; until April 2021, the FDA had never publicly threatened to enforce the \$10,000-a-day civil penalty. This changed when it issued its first non-compliance notice to Acceleron Pharma, and the company then posted its results on ClinicalTrials.gov.

## 2.2. Competitor Priority Review Drug Approvals

Changes in a firm’s competitive landscape and its competitive expectations can occur at any stage of the drug development process, such as following a competitor’s

initiation of clinical trial testing. However, a salient shift in a firm’s competitive environment occurs when a competitor’s drug gains regulatory approval. The regulatory approval of a competitor’s drug resolves uncertainty about the competitor’s ability to market the drug for a specific use and also serves as a type of regulatory quality certification that highlights the drug’s potential value to the market (Dranove and Jin 2010). Following Gilchrist (2016), we focus on important competitor drug approvals: approvals of competitor drugs granted priority review status by the FDA.

The process by which a firm receives approval for a drug with priority review begins with a firm submitting an NDA (for more details, see U.S. Food and Drug Administration 2007). Upon receipt, the FDA must evaluate whether the clinical evidence in the NDA justifies a priority review designation, which the firm can specifically request. The FDA grants priority review designation to drugs found to be exceptionally safe or effective. Notably, the minimum level of safety and efficacy evidence required for drugs with nonpriority (standard) and those with priority review is the same.

Importantly, a drug’s priority review approval raises its profile and has important financial implications for its manufacturer.<sup>10</sup> For example, Aryal et al. (2023) use stock market event studies to show that the announcement of a firm’s drug approval leads to significant increases in that firm’s market value.<sup>11</sup>

## 2.3. Conceptual Framework

This section motivates the empirical exercise with a discussion of the literature on competitive dynamics and disclosure, which are then applied to clarify how pharmaceutical firms shift their disclosure decisions in response to competitor priority review drug approvals.

A growing literature has yielded insight into how competitors’ success in the same product market (disease) can shape their rivals’ decisions. The approval of a competitor’s priority review drug in the same product market in which a firm is conducting its own clinical trial may shift the firm’s competitive expectations, raising its concerns about negative impact on its value because of dampening consumer interest (see, e.g., Tirole 1988; Davis 2006; Jansen 2010, 2017; McCann and Vroom 2010; Bloom et al. 2013; Gentzkow and Kamenica 2017; Cookson 2018).<sup>12</sup> The intuition behind this is fairly straightforward; the approval of a competitor drug in the same market is the regulatory certification of an effective treatment option for consumers in that market. For the firm, the availability of an alternative (potentially, safer and more effective) option among customers increases the risk of having reduced market power, profit margins, and/or market share.<sup>13</sup> These effects are amplified by the possibility that the competitor may benefit from first-mover advantages (e.g., because of high patient switching costs).<sup>14</sup> A competitor’s priority

review drug approval may also reveal differences in the firm's underlying R&D capabilities (Teece and Pisano 2003). Consistent with this view, the approval of a drug has been shown to lower the market value of competing firms with drugs in the same product market (Laise 2023). In addition, studies show that more drugs in the same product market can result in lower average drug prices (see, e.g., Lu and Comanor 1998 and Darrow and Kesselheim 2018).

As a result, a competitor's priority review drug approval in the same product market in which a firm is conducting its own clinical trial may raise the firm's concerns about dampening consumer interest and increase the strategic significance of the information generated from that trial. In response to such an approval, a firm may respond by withholding information on its R&D projects. For example, faced with a greater threat of business stealing by rivals, firms may be less likely to disclose information from failed projects suggesting lower quality. Firms may be concerned that such disclosures could lead consumers to view them negatively compared with newly successful substitutes, thus dampening demand for its future (for firms developing novel drugs) and existing (for firms developing new uses of already approved drugs) products. Therefore, a competitor's priority review drug may cause firms testing drugs in the same disease to reduce and delay their disclosure of results.

In addition to being shaped by consumers' potential responses, firms' decisions to disclose trial results may also be influenced by their competitors' potential use of it (Arrow 1972). Note importantly that firms with newly approved drugs that are particularly promising (i.e., those with priority review) have strong incentives to continue their R&D efforts. Given the high costs of developing novel drugs, such firms are likely to pursue additional approvals for use in new patient groups, therefore expanding the drug's market size. Indeed, 67% of drugs initially approved with priority review go on to expand their label to include additional indications or trial results.<sup>15</sup> Existing drug regulations strengthen these incentives by permitting manufacturers of approved drugs that are seeking supplemental approvals to expedite the R&D process; such firms may bypass early R&D stages (because of the drug's known safety profile) and may require fewer phase II trials (U.S. Food and Drug Administration 1998).

As a result, firms may withhold results to avoid negatively affecting consumers' perceptions—likely a first-order effect (Viscusi 1978, Grossman and Hart 1980, Jovanovic 1982)—and also, to prevent competitors with approved drugs from quickly utilizing the information in their own future R&D efforts. These competitors, who are already at an advantage as they may now be able to expedite the R&D process, might find such disclosures valuable. For example, a competitor with a

priority review drug approved for postmenopausal breast cancer might aim to also get it approved for premenopausal patients. Another firm conducting a successful trial on a breast cancer drug for all age groups might withhold results from successful and failed trials to avoid helping its competitors' R&D efforts. Such results could provide competitors insights into potential risks (e.g., adverse events in premenopausal patients), new research avenues (e.g., additional patient subgroups), and probabilities of success (e.g., the likelihood of demonstrating efficacy in premenopausal patients) in a particular disease market.<sup>16</sup> Highlighting concerns around free riding, the Pharmaceutical and Research Manufacturers of America and the European Federation of Pharmaceutical Industries and Associations, the leading pharmaceutical industry associations, noted in their clinical trial data-sharing guidance that “one of the risks to innovation is disclosure to competitors of companies' trade secrets and proprietary information that could allow others to ‘free ride’ off of the substantial investments of innovators” (Pharmaceutical Research and Manufacturers of America (PhRMA) and European Federation of Pharmaceutical Industries and Associations (EFPIA) 2023). Therefore, a competitor's priority review drug may cause firms testing drugs in the same disease to withhold their disclosure of results of both failed and successful projects.

Finally, a growing literature suggests that although competitor success can lead to shifts in firms' competitive expectations, raising their concerns about the likelihood of dampening consumer interest, in some cases, the negative effects of competitor entry may be offset by the benefits (see, e.g., Davis 2006; Jansen 2010, 2017; Gentzkow and Kamenica 2017). These benefits are likely to be salient when the competitor's product is technologically similar (Bloom et al. 2013); in highly uncertain settings, the regulatory approval of a technologically similar competitor product acts as a certification of the product's technological features. Such an approval can provide valuable information to consumers about the clinical value of drugs with similar technologies, thus creating positive spillover effects for such drugs. These effects may counteract concerns about the negative effects typically associated with competitor drug approvals (e.g., dampening consumer interest and the leakage of valuable information to competitors). As a result, a laggard with a drug that is technologically similar to a recently approved rival drug may decide to disclose its results to signal to consumers that its product is the same as (or better than) its rival's newly successful drug.

Note importantly that the success of a technologically similar product can further counteract the negative effects typically associated with competitor drug approvals by revealing valuable information about the likely regulatory or market success of a firm's product. Additionally, lagging firms may benefit by relying on

investments in consumer education and infrastructure made by early movers with technologically similar products (Lieberman and Montgomery 1988). Therefore, the impact of competitor drug approvals on firms' disclosure decisions may vary across competitor type. Although we expect that competitor approvals typically discourage disclosure among rivals, this effect may be lessened (and even reversed if the positive spillover effects are sufficiently strong) when the competitor's drug is technologically similar.

Taken together, this body of research and this empirical setting support our main prediction; on average, a competitor's priority review drug approval will cause firms testing drugs in the same disease to reduce and delay their disclosure of results. We directly investigate this prediction in our main empirical analysis. In subsequent sections, we further examine how the relationship between competitor approvals and disclosure varies across failed and successful projects, competitor type, and other important sources of heterogeneity.

### 3. Data

To understand the relationship between competitor priority review drug approvals and disclosure, we begin with data on clinical trials registered on ClinicalTrials.gov, which we obtain from the Aggregate Analysis of ClinicalTrials.gov Database (Clinical Trials Transformation Initiative 2024). For each clinical trial, we obtained data on the tested drug (e.g., Olaparib), sponsoring firm (e.g., AstraZeneca), sponsor type (e.g., private sector), trial start date (as determined by the date on which the first patient was enrolled), phase, and clinical trial duration.

Following recent work (see, e.g., Allain et al. 2016, Krieger 2021, Krieger et al. 2022a), our primary market categorization comes from identifying the disease that each trial is treating (e.g., breast cancer). ClinicalTrials.gov categorizes trial diseases using Medical Subject Heading (MeSH) codes, a hierarchical controlled vocabulary maintained by the National Library of Medicine. The set of MeSH codes linked to each trial is publicly available. Using nine-digit MeSH codes to delineate discrete product markets (e.g., eye neoplasms; pneumonia, viral), we generated a trial-MeSH-level data set, where we related a trial's results reporting to competitor drug approvals within that MeSH code.<sup>17</sup>

We next incorporated data from Clarivate Analytics' Cortellis Competitive Intelligence and Clarivate Analytics' Cortellis Clinical Trials Intelligence (hereafter, "Cortellis") to incorporate additional trial characteristics (e.g., the molecular mechanism of action). We also proxied for the costs of disclosure (i.e., the marginal costs associated with collecting, preparing, analyzing, and submitting trial data) by collecting data on clinical trial participant enrollment and the use of clinical trial services (i.e., contract research organizations) (Mello et al. 2013).<sup>18</sup>

As a proxy for the market size of the trial disease (*Disease Market Size*), we obtained estimates of national patient diagnoses using the Medical Expenditure Panel Survey database, counting the total number of patients diagnosed in each trial's associated diseases in the year that the trial began.<sup>19</sup> As a proxy for sponsor R&D experience (*Sponsor Experience*), we counted the total number of clinical trials initiated by the trial's sponsor within the previous five years.

Finally, we make two sample restrictions. First, our sample is limited to phases II, II/III, and III for four reasons. Nearly all nonphase I trials are required to be registered on ClinicalTrials.gov, they constitute a substantial investment (Martin et al. 2017 report that median phase II and phase III costs are \$8.6 million and \$21.4 million, respectively), they assess both critical safety and efficacy outcomes, and they are the focus of clinical trial results reporting regulations (U.S. Food and Drug Administration 2007).<sup>20</sup> Second, our sample is limited to trials that started in or after 2007 and were completed in or by 2019 (to allow for at least two years of results reporting). This results in a final sample of 147,413 clinical trial-MeSH observations.

Next, we collected data on clinical trial results disclosure, competitor priority review drug approvals, project success, and reported clinical outcomes.

#### 3.1. Clinical Trial Results Disclosure

Our primary measures of disclosure come from examining clinical trial results reporting on ClinicalTrials.gov. We focus on two key measures: (i) an indicator for whether results were reported within two years of clinical trial completion and (ii) if results were reported, the time from trial completion to results reporting.

#### 3.2. Competitor Drug Approvals

Data on competitor priority review drug approvals come from Drugs@FDA and the Compilation of Center for Drug Evaluation and Research New Molecular Entity Drug and New Biologic Approvals data set. For each trial-MeSH observation, we identify the earliest date of a competitor priority review FDA drug approval within the same MeSH code. Indications associated with each drug approval were mapped to MeSH codes using manual lookups, the Clinical Drug Experience Knowledgebase, and the public assessment reports from the European Medicines Agency. By 2020, 29% of trial-MeSH observations experience a competitor priority review drug approval in the same MeSH code.

#### 3.3. Project Success and Reported Clinical Outcomes

We next construct a measure of project quality for all trials, regardless of whether their results are reported. Determining the quality of all R&D projects is inherently challenging because it can depend on a variety of complex, unobservable, difficult-to-measure factors, such as meaningful changes in clinical trial end points



(outcomes). In keeping with the heterogeneity available in our context, we utilize the fact that in the pharmaceutical setting, firms are more likely to consider an R&D project successful and to advance it to the next stage in the development and approval process if the project is considered sufficiently high quality (i.e., the tested drug has been shown to be sufficiently safe and effective for a specific indication) (Krieger 2021, Garfinkel and Hammoudeh 2024). Thus, in a subset of the empirical analyses that follow, we examine project success as proxied by whether a phase II trial advances to phase III or by whether a drug from a phase III trial gains approval.<sup>21</sup> Note importantly that one caveat to interpreting results that incorporate project advancement measures is that such measures are not a perfect proxy for project quality. Although trial advancement may be broadly correlated with a project's quality, projects can progress for many reasons other than drug safety and efficacy, such as firms' progression-seeking incentives, managerial biases, broader economic conditions, and varying interpretations of the results (Guedj and Scharfstein 2004, Cook et al. 2014, Lendrem et al. 2015). This suggests that a firm that just monitors its competitor's project progression cannot fully capture the true quality of its rival's projects. This highlights the importance of the information gained from the results of disclosed trials for external stakeholders.

In addition to characterizing trials by their advancement status, we characterize trials based on whether the reported clinical outcomes were favorable. We categorize trials as having favorable reported clinical outcomes using two types of information. First, for all trials, a favorable reported clinical outcome is indicated by a statistically significant ( $p < 0.05$ ) change in the treatment group clinical end points (e.g., the average five-year survival, a cardiac event). Second, for cancer trials, we consider the objective response rate to treatment, a commonly used clinical trial end point that describes the percentage of participants whose tumors were reduced by a prespecified amount.<sup>22</sup>

Table 1 presents baseline summary statistics on our trial-MeSH level data set. Between 2007 and 2019, roughly 34% of trial-MeSH observations had reported results on ClinicalTrials.gov within two years of clinical trial completion. Figure B1 in the Online Appendix shows that only 20% of trial-MeSH observations report results within the FDAAA-mandated time frame of one year. Figure B2 in the Online Appendix shows that although one-year reporting rates increased between 2000 and 2016, when they just exceeded 25%, they have been declining in recent years. Additionally, Table 1 shows that 29% of trial-MeSH observations face a competitor priority review drug approval in the same MeSH code.

## 4. Descriptive Facts About Disclosure and Competitor Activity

Before turning to the main causal analysis, this section provides a descriptive analysis of disclosure practices and competitor activity in the pharmaceutical industry. We first probe one of the paper's assumptions: that firms are selectively disclosing trial results. We then show that consistent with the conceptual framework's key prediction, competitor activity likely shapes firms' disclosure practices.

### 4.1. Facts About Disclosure

Using proxies for project quality and data on reported clinical outcomes, we document two patterns that shed light on firms' disclosure decisions. First, Table B1 in the Online Appendix shows that, on average, trials that successfully advance are 14% ( $\approx \frac{-0.054}{0.382} \cdot 100$ ) more likely to have disclosed results within two years following trial completion relative to those that did not. This corroborates the view that firms are likely to disclose results above a quality threshold.<sup>23</sup> The results in Table B1 in the Online Appendix document the relationship between quality and disclosure, averaged across firms that are facing competitor drug approvals and firms that are not facing competitor drug approvals. In subsequent sections, we find that although quality is correlated with

**Table 1.** Summary Statistics

| Variable                                                               | Mean     | Median   | Standard deviation | Min | Max     |
|------------------------------------------------------------------------|----------|----------|--------------------|-----|---------|
| 0/1: Results reported on ClinicalTrials.gov                            | 0.34     | 0.00     | 0.47               | 0   | 1       |
| Months to results reporting on ClinicalTrials.gov                      | 26.24    | 16.95    | 23.11              | 0   | 163     |
| 0/1: Competitor priority review drug approval in MeSH                  | 0.29     | 0.00     | 0.45               | 0   | 1       |
| Disease market size (no. of diagnoses in disease, 1,000s)              | 7,913.74 | 2,610.02 | 11,034.21          | 11  | 68,497  |
| Sponsor experience (no. clinical trials, past 5 years)                 | 1.85     | 0.00     | 4.44               | 0   | 51      |
| No. of patients                                                        | 304.57   | 84.00    | 1,589.22           | 0   | 204,438 |
| Trial length in months                                                 | 27.12    | 21.97    | 20.21              | 0   | 143     |
| 0/1: Phase 2 trial                                                     | 0.55     | 1.00     | 0.50               | 0   | 1       |
| 0/1: Phase 3 trial                                                     | 0.45     | 0.00     | 0.50               | 0   | 1       |
| 0/1: Successful trial                                                  | 0.35     | 0.00     | 0.48               | 0   | 1       |
| 0/1: Disclosed clinical outcome – significant outcome                  | 0.63     | 1.00     | 0.48               | 0   | 1       |
| Disclosed clinical outcome for cancer trials – objective response rate | 28.48    | 21.85    | 26.21              | 0   | 100     |

*Notes.* This table shows summary statistics at the trial-MeSH code level. For detailed variable descriptions, see Section 3.



disclosure, competitor drug approvals play a distinct role in shifting whether and when firms report trial results.

Second, highlighting the selective nature of firms' disclosure decisions, Figure B3 in the Online Appendix reveals that firms tend to disclose more favorable trial results soon after trial completion while delaying the release of less favorable trial results, even after accounting for key trial, firm, and market characteristics. For example, panel (c) of Figure B3 in the Online Appendix suggests that firms that reported their objective response rates after a one-year delay had rates that were, on average, two percentage points lower than those that reported immediately after trial completion. These two new facts suggest that firms are selective about what and when they disclose trial results. We now focus on how firms' disclosure decisions may relate to changes in their competitive landscape.

#### 4.2. Facts About Competitor Activity

Before turning to a causal investigation of competitor priority review drug approvals on firms' disclosure decisions, we begin by providing descriptive evidence about the relationship between competitive dynamics and disclosure. In keeping with a large literature that characterizes a firm's competitive landscape by using counts of competitor trial activity within the same product market (see, e.g., Cunningham et al. 2021 and Krieger 2021), we count the number of phase II competitor trials initiated in the same MeSH code within the five years prior to each focal trial-MeSH observation.<sup>24</sup>

For a given trial-MeSH observation, this approach identifies competitors using similarly advanced drug development approaches as the focal trial.

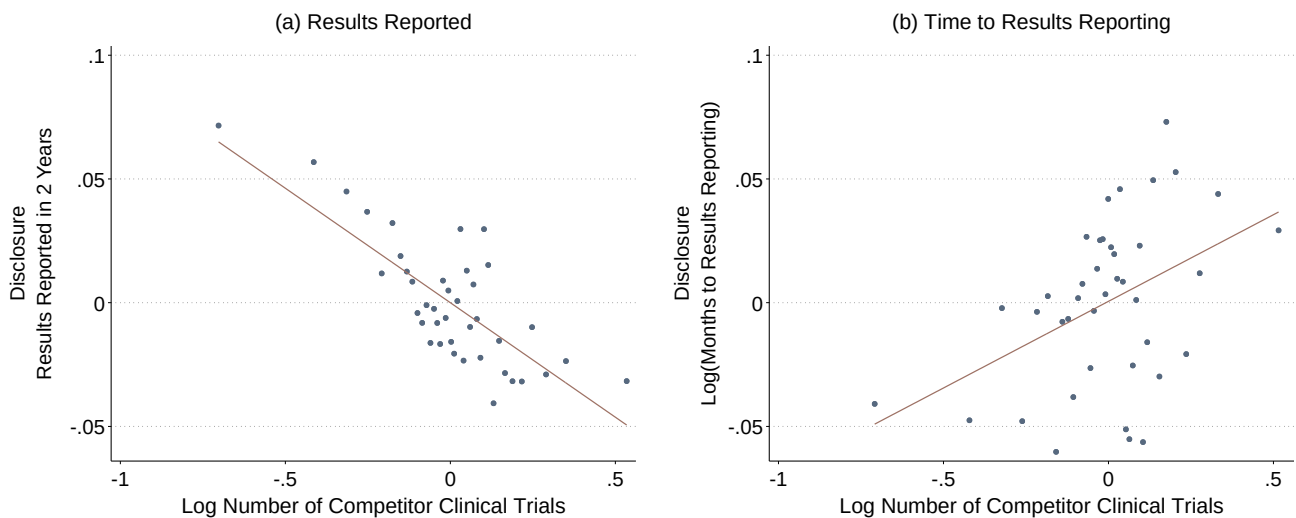
Using these data, we examine the relationship between the number of phase II competitor trials and firms' disclosure decisions. For trial-MeSH observation  $id$ , we estimate the following:

$$Disclosure_{id} = \beta_1 CompetitorTrials_{id} + \gamma' X_{id} + \delta_d + \epsilon_{id}, \quad (1)$$

where  $Disclosure_{id}$  measures clinical trial results reporting and  $CompetitorTrials_{id}$  is the log of one plus the number of private- and public-sector phase II competitor trials initiated in MeSH code  $d$  within five years prior to the start of clinical trial  $i$ . Our main regressions include disease (MeSH code) fixed effects ( $\delta_d$ ) and detailed clinical trial controls ( $X_i$ ) for disease market size; sponsor experience; and trial participant enrollment, duration, phase, and start year fixed effects.<sup>25</sup> This allows us to examine the relationship between the level of competitor trials and disclosure, *net* of factors such as differences in scientific opportunities, market demand, disclosure costs, and firm resources.<sup>26</sup> We estimate ordinary least squares (OLS) linear probability models with heteroskedasticity-robust standard errors clustered at the MeSH code.<sup>27</sup>

Figure 1 and Table 2 show that consistent with our conceptual framework, differences in firms' disclosure decisions are related to differences in their competitive landscape. Column (3) in Table 2 suggests that a 10%

Figure 1. (Color online) Competitor Clinical Trials and Disclosure



Notes. This figure shows the relationship between the number of competitor clinical trials and trial results reporting on ClinicalTrials.gov. Observation is at the trial-MeSH level. The sample consists of trials initiated in/after 2007 and completed by 2019. Competitor trials for a trial-MeSH pair are the log of the number of competitor phase II trials initiated in the same MeSH code within the previous five years. Panel (a) shows a bin scatterplot comparing the likelihood of results reporting on the site within two years of the trial completion date against the trial-MeSH's number of competitor trials, and panel (b) shows for reported trials the time to results reporting from the trial completion date against the trial-MeSH's number of competitor trials. Variables are residualized on year fixed effects; MeSH code fixed effects; sponsor experience; trial phase, duration, and the number of patients enrolled; and the number of patients diagnosed with the trial's diseases in the trial's start year. For more detailed variable descriptions, see Section 3.

Table 2. Competitor Clinical Trials and Disclosure

| Dependent variable:        | Results Reported       |                        |                         | log(Months to Reporting) |                       |                         |
|----------------------------|------------------------|------------------------|-------------------------|--------------------------|-----------------------|-------------------------|
|                            | (1)                    | (2)                    | (3)                     | (4)                      | (5)                   | (6)                     |
| log(Competitor Trials)     | −0.119***<br>(0.00859) | −0.120***<br>(0.00864) | −0.0925***<br>(0.00848) | 0.0618***<br>(0.0178)    | 0.0815***<br>(0.0180) | 0.0734***<br>(0.0185)   |
| log(Disease Market Size)   |                        | 0.00445**<br>(0.00186) | 0.00131<br>(0.00142)    |                          | 0.00468<br>(0.00328)  | −0.0118***<br>(0.00279) |
| log(Sponsor Experience)    |                        |                        | 0.158***<br>(0.00309)   |                          |                       | −0.0582***<br>(0.00609) |
| Mean of dependent variable | 0.336                  | 0.336                  | 0.336                   | 3.025                    | 3.025                 | 3.025                   |
| Observations               | 147,283                | 147,283                | 147,283                 | 77,991                   | 77,991                | 77,991                  |
| R <sup>2</sup>             | 0.062                  | 0.084                  | 0.152                   | 0.118                    | 0.138                 | 0.183                   |
| Start year FE              | Yes                    | Yes                    | Yes                     | Yes                      | Yes                   | Yes                     |
| MeSH code FE               | Yes                    | Yes                    | Yes                     | Yes                      | Yes                   | Yes                     |
| Trial controls             | No                     | Yes                    | Yes                     | No                       | Yes                   | Yes                     |

Notes. This table shows the relationship between the number of competitor trials and results reporting on ClinicalTrials.gov. Observation is at the trial-MeSH level. The sample consists of trials initiated in/after 2007 and completed by 2019. *Competitor Trials* for a trial-MeSH is the number of competitor phase II trials initiated in the same MeSH code within the previous five years. Estimates are from OLS models. Columns (1)–(3) regress an indicator for trial results reporting on the site within two years of trial completion. For trials that do report, columns (4)–(6) regress the log of the time from trial completion to reporting. *Disease Market Size* denotes the number of patients diagnosed with the trial’s diseases in the trial start year. *Sponsor Experience* denotes the number of trials initiated by the trial sponsor in the same MeSH code within the previous five years. *Trial Controls* denotes controls for trial phase, trial duration, and the number of participants enrolled. All columns include year and MeSH code fixed effects. Robust standard errors, clustered at the MeSH code level, are shown in parentheses. Singleton observations are dropped in fixed effects specifications, which accounts for the smaller number of observations. For detailed variable descriptions, see Section 3.

\* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

increase in the level of phase II competitor trials is associated with a 2.8% ( $\approx 10 \cdot \frac{-0.093}{100} \cdot \frac{1}{0.336} \cdot 100$ ) decrease in the probability of a firm reporting results on ClinicalTrials.gov within two years following trial completion. Additionally, higher levels of competitor trials are associated with delays in results reporting. On average, among trials that reported results, a 10% increase in the level of phase II competitor trials was associated with a roughly 0.7% increase in time to results reporting.

Online Appendix C provides a detailed discussion of robustness checks and extensions on these correlations. We present a summary of our six main findings. First, we confirm that the competitor activity-disclosure relationship is robust to alternative measures of competitor activity (e.g., those based on phase III trials, expected drug approvals, actual drug approvals, and firms) (see Table C1 in the Online Appendix). Second, corroborating the view that private-sector institutions are more focused on profits and more responsive to competitor trials relative to public-sector institutions (see, e.g., David et al. 2000), we observe significantly stronger effects in private-sector trials (Table C2 in the Online Appendix). Third, Table C4 in the Online Appendix shows that our results are robust to using alternative strategies for linking trials to diseases and alternative disease measures (e.g., International Classification of Diseases, 9th Edition (ICD-9) codes). Fourth, we show that our findings are robust to other dimensions of disclosure, such as reporting in publications and conferences (Table C5 in the Online Appendix). Fifth, Table

C6 in the Online Appendix shows our results are robust when controlling for contract research organizations, which may change sponsors’ marginal trial reporting costs (Mirowski and Van Horn 2005). Finally, to demonstrate that the estimated competitor trial-disclosure relationship is unlikely to be statistical noise, we run 2,000 placebo experiments where trial-based competitor trial levels are randomized across trial-MeSH observations (Figure C2 in the Online Appendix).

5. Effects of Competitor Approval on Disclosure

5.1. Empirical Strategy

Section 4 documents what is, to the best of our knowledge, a new fact: that the level of competitor activity—as measured by competitor trials—is strongly negatively correlated with disclosure. However, concerns around unobservable factors (e.g., heterogeneity in demand or scientific opportunities) may limit the causal interpretation of the results. Although regression controls (e.g., disease fixed effects) mitigate some of these concerns, unobservable factors may continue to drive these correlations. In the following analyses, we address these concerns by using uncertainty in the drug approval process to examine how a plausibly exogenous change in the competitive landscape, in the form of competitor priority review drug approvals, can alter firms’ competitive expectations and influence their disclosure decisions. Variation in the timing of competitor drug approval

shocks allows not-yet-treated trial-MeSH observations<sup>28</sup> to serve as a plausible control group.<sup>29</sup>

A key assumption underlying this analysis is that in the years prior to a competitor's priority review drug approval, firms cannot fully anticipate if and when this might happen. In practice, there is substantial public uncertainty regarding the date on which a priority review drug might receive approval and other events preceding this, such as submission of the NDA and FDA assignment of priority review status (Morton 1999, Rao 2022).<sup>30</sup> We provide three pieces of evidence suggesting significant industry uncertainty about the exact timing of such approvals and that such approvals signal a shift in stakeholder expectations. First, there is substantial variation in the approval timing; the average regulatory review period for priority review drugs is 227 days, with a standard deviation of 79 days. Second, as we describe below, we conduct an event study that explores the impact of competitor priority review drug approval on disclosure. An analysis of pretrends in the event study yields little support for anticipatory effects (Figure 2). Third, stock market event studies that document changes in a drug firm's stock prices around the announcement of an FDA approval suggest an economically significant increase in firm market value following the announcement and little information leakage prior to it (Aryal et al. 2023).<sup>31</sup>

To test the effect of competitor priority drug approvals on results reporting, we estimate

$$\text{Disclosure}_{id} = \beta_1 \text{PostCompetitorDrugApproval}_{id} + \gamma' X_{id} + \delta_d + \epsilon_{id}, \quad (2)$$

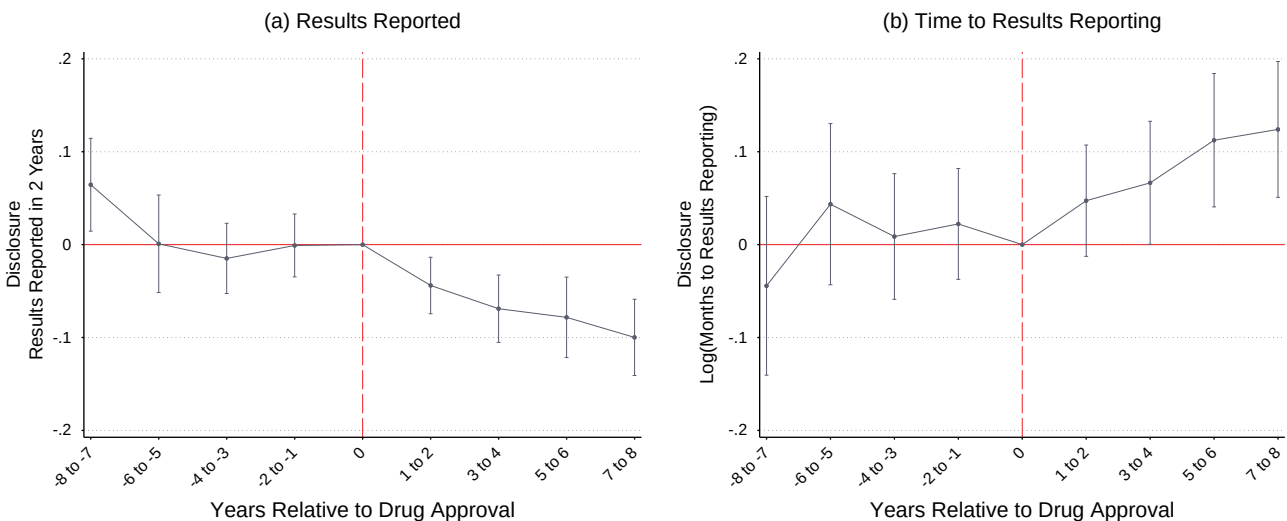
where  $\text{PostCompetitorDrugApproval}_{id}$  is an indicator for whether a competitor priority review drug approval has occurred in MeSH code  $d$  by the start of clinical trial  $i$ .<sup>32</sup> Importantly, our analysis focuses on the impact of competitor drug approvals at the year level. This allows us to account for regulatory events that may be closely related to the timing of the FDA approval of a competitor's priority review drug, such as the initial NDA submission and the subsequent priority review designation.<sup>33</sup>

The remaining variables are the same as in Equation (1). Our coefficient of interest is  $\beta_1$ , which represents the causal effect of a competitor priority review drug approval on disclosure within the same MeSH code. In particular,  $\beta_1$  measures the average difference in disclosure in treated and control trial-MeSH observations before and after such an approval, conditional on controls. We focus on the set of such approvals made on and after 2005 (i.e., within two years of the start of our trial sample) as we want to consider the effects of competitor drug approvals that occur even prior to the start of our earliest trial (in 2007). Following the trial funding results shown in Table C2 in the Online Appendix, our analysis centers on private-sector trials, which are more likely to respond to shifts in the competitive landscape.

## 5.2. Baseline Results

Columns (1)–(3) in Table 3 show that across all private-sector trials, a competitor priority review drug approval decreases clinical trial results reporting. In particular, such an approval decreases the probability of results

Figure 2. (Color online) Event Study Estimates: Impact of Competitor Drug Approvals



Notes. This figure plots the change in ClinicalTrials.gov trial results reporting following a competitor priority review drug approval. In panel (a), the outcome is a binary indicator for whether trial results are reported on the site within two years of trial completion. In panel (b), for reported trials, the outcome is the log of the time to results reporting. The specification is based on trial-MeSH observations, and the sample consists of private-sector trials initiated in/after 2007 and completed by 2019. OLS regressions include year fixed effects; MeSH code fixed effects; controls for trial phase, duration, and the number of participants enrolled; disease market size; and sponsor experience. For more detailed variable descriptions, see Section 3.



**Table 3.** Impact of Competitor Priority Drug Approvals on Disclosure

| Dependent variable:                  | Results Reported       |                        |                         | log(Months to Reporting) |                        |                         |
|--------------------------------------|------------------------|------------------------|-------------------------|--------------------------|------------------------|-------------------------|
|                                      | (1)                    | (2)                    | (3)                     | (4)                      | (5)                    | (6)                     |
| <i>Post Competitor Drug Approval</i> | −0.0543***<br>(0.0193) | −0.0527***<br>(0.0192) | −0.0528***<br>(0.0142)  | 0.0421*<br>(0.0254)      | 0.0452*<br>(0.0254)    | 0.0474**<br>(0.0223)    |
| <i>log(Disease Market Size)</i>      |                        | 0.000416<br>(0.00218)  | −0.00454**<br>(0.00190) |                          | −0.000157<br>(0.00418) | 0.00393<br>(0.00396)    |
| <i>log(Sponsor Experience)</i>       |                        |                        | 0.131***<br>(0.00317)   |                          |                        | −0.0761***<br>(0.00696) |
| Mean of dependent variable           | 0.408                  | 0.408                  | 0.408                   | 3.058                    | 3.058                  | 3.058                   |
| Observations                         | 78,781                 | 78,781                 | 78,781                  | 51,073                   | 51,073                 | 51,073                  |
| R <sup>2</sup>                       | 0.132                  | 0.144                  | 0.187                   | 0.178                    | 0.205                  | 0.211                   |
| Start year FE                        | Yes                    | Yes                    | Yes                     | Yes                      | Yes                    | Yes                     |
| MeSH code FE                         | Yes                    | Yes                    | Yes                     | Yes                      | Yes                    | Yes                     |
| Trial controls                       | No                     | Yes                    | Yes                     | No                       | Yes                    | Yes                     |

*Notes.* This table reports DiD estimates of the effect of competitor priority review drug approvals on results reporting on ClinicalTrials.gov. Observation is at the trial-MeSH level. The sample consists of private-sector trials initiated in/after 2007 and completed by 2019. Estimates are from OLS models. Columns (1)–(3) regress an indicator for trial results reporting on the site within two years of trial completion. For trials that do report, columns (4)–(6) regress the log of the time from trial completion to reporting. *Post Competitor Drug Approval* switches from zero to one when a competitor drug is approved in the same MeSH code as the focal trial-MeSH. *Disease Market Size* denotes the number of patients diagnosed with the trial’s diseases in the trial start year. *Sponsor Experience* denotes the number of trials initiated by the trial sponsor in the same MeSH code within the previous five years. *Trial Controls* denotes controls for trial phase, trial duration, and the number of participants enrolled. All columns include year fixed effects and MeSH code fixed effects. Robust standard errors, clustered at the MeSH code level, are shown in parentheses. Singleton observations are dropped in fixed effects specifications, which accounts for the smaller number of observations. For more detailed variable descriptions, see Section 3.

\* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

reporting by roughly 13% ( $\approx \frac{-0.053}{0.41} \cdot 100$ ). Columns (4)–(6) in Table 3 indicate that if a firm does report results, a competitor priority review drug approval delays this reporting by roughly 5%. Controlling for market size and additional trial characteristics does little to change these estimates.

To explore the timing of the estimated impact on the probability of results reporting, we estimate an event study version of Equation (2), where the single indicator variable  $PostCompetitorDrugApproval_i$  is replaced with a vector of indicator variables for two-year bins before and after the first competitor priority review drug approval in the focal MeSH code. Providing support for our identification strategy, panel (a) of Figure 2 shows that the probability of results reporting does not change differentially across trial-MeSH observations before the approval. However, the probability of reporting declines differentially for trial-MeSH observations that experience a competitor priority review drug approval and remains lower afterward. Firms do shift disclosure decisions relatively quickly; firms lower their results reporting within two years of a competitor drug approval. Consistent with these results, panel (b) of Figure 2 shows that a competitor priority review drug approval leads to significant delays in the time to results reporting among disclosing clinical trials.

5.3. Robustness Checks

We probe the robustness of these DiD estimates in six ways. Table D1 in the Online Appendix confirms that

these findings are robust: first, to the use of alternative disease categorizations (such as ICD-9 codes) and second, after adding additional controls for marginal disclosure costs (contract research organization involvement). Third, we ask whether our results are robust to restricting the set of trial-MeSH observations to just those associated with disease types associated with priority review. Table D1 in the Online Appendix shows that results are robust to restricting the analysis to the set of diseases that the FDA considers “serious” or “life threatening.”<sup>34</sup> Fourth, we investigate the robustness of our results to various treatment windows. This is motivated by concern that a competitor drug approval may not impact the disclosure decisions for trials initiated many years postapproval. Our main analysis focuses on eight-year windows as we expect that a competitor drug approval may have long-term effects, but Table D1 in the Online Appendix confirms that our results are robust to restricting the treatment window to two, three, and four years before and after the time of the competitor priority review approval. Fifth, Figure D1 in the Online Appendix investigates concerns about MeSH-level selection by comparing a proxy for research and market potential—publications—among MeSH codes with early competitor priority review drug approvals (i.e., prior to the median drug approval date) between 1990 and 2006 and those with late or no approvals. Although publication levels differ, they do not change substantially over time, suggesting that the inclusion of MeSH code fixed effects in our regressions

minimizes the risk of bias because of MeSH-level selection.

Our sixth and last robustness check centers around the recent DiD literature, which highlights that treatment that is staggered over time generates a weighted average of all possible pair-wise DiD estimates. As a result, already-treated observations are used as a control for late-treated observations, resulting in negative weights in calculating the average treatment effect (de Chaisemartin and D'Haultfoeuille 2020, Sun and Abraham 2021, Borusyak et al. 2024). In addition to estimating event study style specifications, we address these concerns in two ways. First, we estimate Equation (2) excluding trial-MeSH observations that are always treated. Second, in the spirit of Callaway and Sant'Anna (2021), Goodman-Bacon (2021), and Prager and Schmitt (2021), we estimate a separate regression for each cohort (where each cohort is the set of trial-MeSH observations that experience a competitor priority review drug approval in the same calendar year) and take the weighted average of the cohort-specific estimates, where each weight is the cohort's share of all treated observations. These estimates are consistent with the baseline drug approval estimates (Table D2 in the Online Appendix).

## 6. Mechanisms and Heterogeneity

In this section, we provide a more detailed examination of how firms alter their disclosure decisions in response to competitor priority review drug approvals. Although it is easier to establish that firms shift their disclosure decisions in response to competitor drug approvals, understanding the mechanisms behind such decisions is challenging as we lack access to the information that firms choose to withhold. However, we leverage institutional features to conduct heterogeneity analyses to shed light on how firms may shift their disclosure decisions as a strategic response to competitor drug approvals. We focus on three factors: project failure and success, competitor type, and firm experience.

### 6.1. Disclosure Across Failed and Successful Projects

The results from the previous section highlight that competitor drug approvals play an important role in shaping their disclosure decisions. The discussion in Section 2 suggests that in response to competitor drug approvals that shift firms' competitive expectations and increase concerns about dampening consumer interest, firms may withhold information from both low- and high-quality projects out of concerns about appearing less favorable to stakeholders, such as consumers and investors, and unintended sharing of useful information to competitors. As described in Section 3, a rough proxy for a project's quality is whether it successfully advances to the next development stage. In

this section, we, therefore, ask the following question. In response to competitor drug approvals, do firms withhold results from projects that fail (i.e., do not advance to the next development stage, suggesting low quality) and those that succeed (i.e., do advance, suggesting high quality)?

The estimates in columns (1) and (2) in Table 4 show that following a competitor's approval, there is a significant and negative impact on results reporting for both failed and successful drug project subsamples, and those in columns (3) and (4) in Table 4 show that firms delay the reporting of results from both project types as well. As expected, the point estimates are greater for failed trials (nearly twice as large when examining time to reporting), although the differences are not statistically significant, likely because of the relatively small sample size of each subsample.<sup>35</sup> Although our main focus is on the effect of competitor approvals on disclosure, as a robustness check, we also examine the competitor trial-disclosure correlation among failed and successful projects. Table C7 in the Online Appendix shows that the negative effect of competitor activity on disclosure primarily occurs among failed projects.

What can we learn from this exercise? We have two takeaways. First, these results are consistent with the idea that in response to competitor drug approvals, firms may withhold results of both low-quality (e.g., to maintain stakeholder interest) and high-quality (e.g., to minimize competitor free riding) projects to maintain their competitive position. The negative effect of competitor approvals on disclosure among high-quality trials is consistent with pharmaceutical firms' view that withholding some results is important for "maintaining incentives for biomedical research." Firms may be concerned that disclosing information may inadvertently help newly successful competitors in their current and future R&D efforts (Pfizer 2019, Pharmaceutical Research and Manufacturers of America (PhRMA) and European Federation of Pharmaceutical Industries and Associations (EFPIA) 2023, Celgene 2024).<sup>36</sup> Second, the fact that firms withhold results from both failed and successful projects implies that external parties (e.g., consumers, competitors, investors) may not be able to completely infer a project's quality solely based on whether it has advanced or not. As we described in Section 3, project advancement, although useful for characterizing the quality of projects with and without results reported, is an imperfect proxy, and advancement may occur for a variety of reasons. This may provide an additional explanation for why the effects among low- and high-quality projects are not statistically significantly different.

### 6.2. Heterogeneity by Competitor Type

Although the main results suggest that competitor drug approvals lead to a decline in disclosure for

**Table 4.** Impact of Competitor Drug Approvals on Disclosure by Project Failure and Success

| Dependent variable:                  | Results Reported       |                          | log(Months to Reporting) |                          |
|--------------------------------------|------------------------|--------------------------|--------------------------|--------------------------|
|                                      | Failed trials<br>(1)   | Successful trials<br>(2) | Failed trials<br>(3)     | Successful trials<br>(4) |
| <i>Post Competitor Drug Approval</i> | −0.0683***<br>(0.0171) | −0.0448*<br>(0.0263)     | 0.103***<br>(0.0268)     | 0.0581*<br>(0.0319)      |
| <i>log(Disease Market Size)</i>      | −0.00396<br>(0.00247)  | −0.00207<br>(0.00334)    | 0.00368<br>(0.00461)     | 0.00540<br>(0.00688)     |
| <i>log(Sponsor Experience)</i>       | 0.130***<br>(0.00388)  | 0.113***<br>(0.00538)    | −0.0649***<br>(0.00835)  | −0.107***<br>(0.00947)   |
| Mean of dependent variable           | 0.386                  | 0.516                    | 3.015                    | 3.084                    |
| Observations                         | 42,710                 | 25,419                   | 25,317                   | 21,070                   |
| R <sup>2</sup>                       | 0.172                  | 0.262                    | 0.192                    | 0.326                    |
| Start year FE                        | Yes                    | Yes                      | Yes                      | Yes                      |
| Mesh code FE                         | Yes                    | Yes                      | Yes                      | Yes                      |
| Trial controls                       | Yes                    | Yes                      | Yes                      | Yes                      |

*Notes.* This table reports that the DiD estimates of the effect of competitor priority review drug approvals on ClinicalTrials.gov clinical trial results reporting vary by R&D project failure and success. Observation is at the trial-MeSH level. The sample consists of private-sector clinical trials in the trial advancement data set that were initiated in/after 2007 and completed by 2019. The algorithm used to construct phase-specific clinical trial advancement measures required us to rely on the set of clinical trials that were mapped to standardized Cortellis drug names. This resulted in a final “trial advancement data set” of 97,438 trial-MeSH observations. Columns (1) and (3) consist of failed trials. Trials testing drugs that do not advance from phase II to phase III or do not win FDA approval are considered failed trials. Columns (2) and (4) consist of trials testing drugs that successfully advanced from phase II to phase III or from phase III to approval. Observation is at the trial-MeSH level. Estimates are from OLS models. All columns regress an indicator for trial results reporting on ClinicalTrials.gov within two years of trial completion. *Post Competitor Drug Approval* switches from zero to one when a competitor drug is approved in the same MeSH code as the focal trial-MeSH. *Disease Market Size* denotes the number of patients diagnosed with the trial’s diseases in the trial start year. *Sponsor Experience* denotes the number of trials initiated by the trial sponsor in the same MeSH code within the previous five years. *Trial Controls* denotes controls for trial phase, trial duration, and the number of participants enrolled. Singleton observations are dropped in fixed effects specifications, which accounts for the smaller number of observations. Robust standard errors, clustered at the MeSH code level, are shown in parentheses. For more detailed data and variable descriptions, see Sections 3 and 6.4.

\* $p < 0.10$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

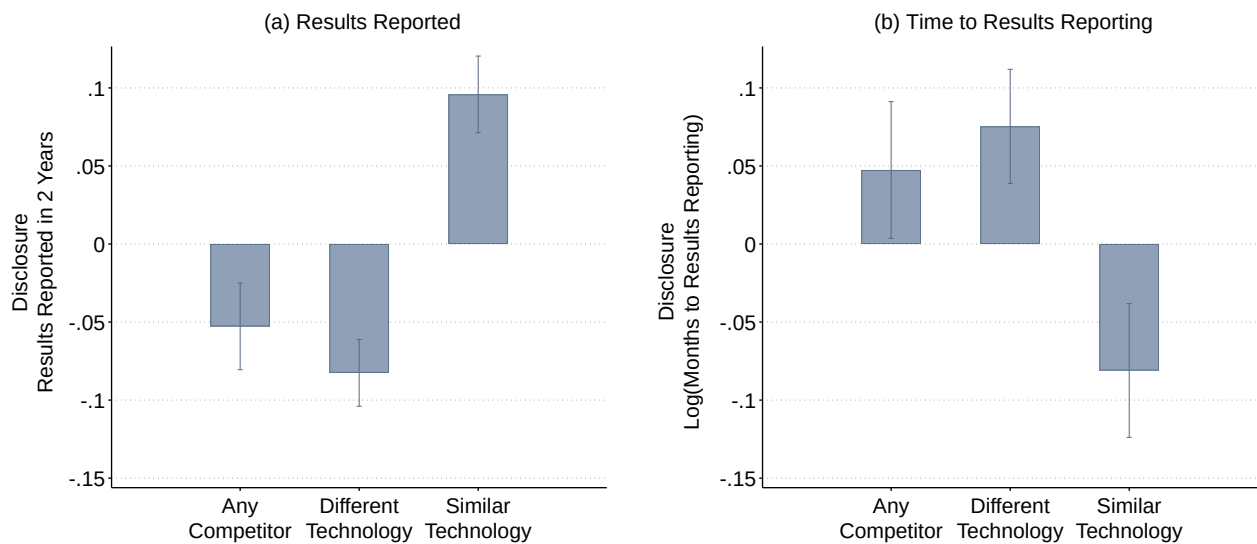
low- and high-quality projects, these averages may mask important differences in the dimensions of competitor drug approvals that shape firms’ disclosure decisions. As discussed in Section 2, the impact of a competitor’s drug approval on a firm’s disclosure decisions can differ. In particular, a firm with a drug that is technologically similar to a successful early entrant may experience positive spillover effects and be eager to disclose its trial results to highlight that its product is comparable with (or better than) the rival drug (Aghion et al. 2005).<sup>37</sup> This contrasts with a situation where the competitor’s product is technologically different; in such a case, the benefits of disclosure may be more limited and outweighed by concerns about dampening consumer interest.

To investigate how the type of competitor approval influences firms’ disclosure decisions, we examine how priority approvals of different types of competitor priority drugs (those based on different and similar technologies) shape firms’ disclosure decisions (Krieger 2021, Markou et al. 2023, Garfinkel and Hammoudeh 2024). Although a drug’s product market refers to the medical condition treated by the drug (e.g., breast cancer), its technology is the biochemical mechanism

through which it achieves its effect (e.g., poly[ADP-ribose] polymerase (PARP) inhibitors produce an anti-cancer effect by preventing damaged cancerous cells from repairing themselves). An individual technology can be associated with multiple medical conditions. For example, PARP inhibitors are used to treat both ovarian cancer and breast cancer.

After incorporating data on the technologies of drugs approved and those tested, we adapt the DiD specification in Equation (2) to separately examine the impact of a competitor priority review drug approval in a different technological area (as a proxy for an approval where the potential for business stealing effects is most salient) or similar technological area (as a proxy for an approval where the potential for positive spillover effects is most salient) as the drug tested in the focal trial. Figure 3 presents results from regressions that are fully saturated with controls that examine the impact on results reporting (panel (a) of Figure 3) and time to results reporting (panel (b) of Figure 3). The first bar in each panel of Figure 3 reproduces the baseline results. We then split the impact of competitor approvals into two mutually exclusive categories—those in a different technology area (the second bar in

**Figure 3.** (Color online) Impact of Competitor Approvals on Disclosure: Heterogeneity by Drug Type



*Notes.* This figure plots the change in results reporting on ClinicalTrials.gov following three types of competitor priority review drug approval in the same MeSH code: (a) any approval, (b) any approval in a different technology area, and (c) any approval in a similar technology area. In panel (a), the outcome is a binary indicator for whether trial results are reported on the site within two years of trial completion. In panel (b), for reported trials, the outcome is the log of the time to results reporting. Shown are 95% confidence intervals. The specification is based on trial-MeSH observations, and the sample consists of private-sector trials initiated in/after 2007 and completed by 2019. OLS regressions include year fixed effects; MeSH code fixed effects; controls for trial phase, duration, and the number of participants enrolled; disease market size; and sponsor experience. For detailed variable descriptions, see Section 3.

each panel of Figure 3) and those in a similar area (the third bar in each panel of Figure 3).

This distinction reveals the nuanced effects of competitor drug approvals on firms' disclosure decisions. Compared with the baseline results for all competitor priority review approvals, firms with such an approval in a different technology area are even more likely to withhold (by 20% versus 13% in the baseline) or if reporting, to delay (by 8% versus 5% in the baseline) reporting (for detailed estimates, see Table E2 in the Online Appendix). However, if the competitor drug approval involves similar technology, firms are *more* likely to disclose their results (by 23%) or if reporting, to *expedite* their reporting times (by 8%). These heterogeneous responses, which were masked in the baseline results (Table 3), imply that most competitor priority review drug approvals occur in different technology areas.<sup>38</sup>

This underlines the importance of understanding how different dimensions of competitor drug approvals shape firms' disclosure decisions; when a rival's drug with a different technology is approved in the same disease market, concerns about dampening consumer interest are more pronounced, causing firms to withhold clinical trial results. However, if the approved drug is based on a similar technology, concerns about business stealing effects are less salient, potentially leading to a decision to disclose as the approval of a technologically similar competitor could benefit a firm

by providing insights into the potential scientific, regulatory, and commercial success of its drugs. A firm may disclose trial results to demonstrate that its drug is comparable with or better than a competitor's recently approved drug. Taken together, these results highlight how firms may selectively withhold negative information and share potentially promising information when faced with a stronger rival.

Recently, the market for weight loss treatments has demonstrated how success for one firm can benefit competitors with similar technologies. The pharmaceutical firms Novo Nordisk and Eli Lilly each produce a glucagon-like peptide-1 (GLP-1) agonist drug: semaglutide and tirzepatide, respectively. Originally approved for diabetes, these blockbuster drugs have gained prominence in recent years as effective weight loss treatments and are undergoing further clinical tests to validate their safety and effectiveness in weight management. Positive clinical outcomes for one firm's GLP-1 drug have led to rising stock prices for both firms (Philippidis 2023). This corroborates the view that the success of one GLP-1 drug highlights the potential success and the expected acceptance of other GLP-1 drugs among patients, physicians, and insurers (Gardner 2023).<sup>39</sup> Notably, the two firms' activities are often closely aligned. For example, Eli Lilly obtained approval of tirzepatide on May 13, 2022, and Novo Nordisk disclosed clinical results for semaglutide just four days later (ACMA 2023).



6.3. Role of Firm Experience

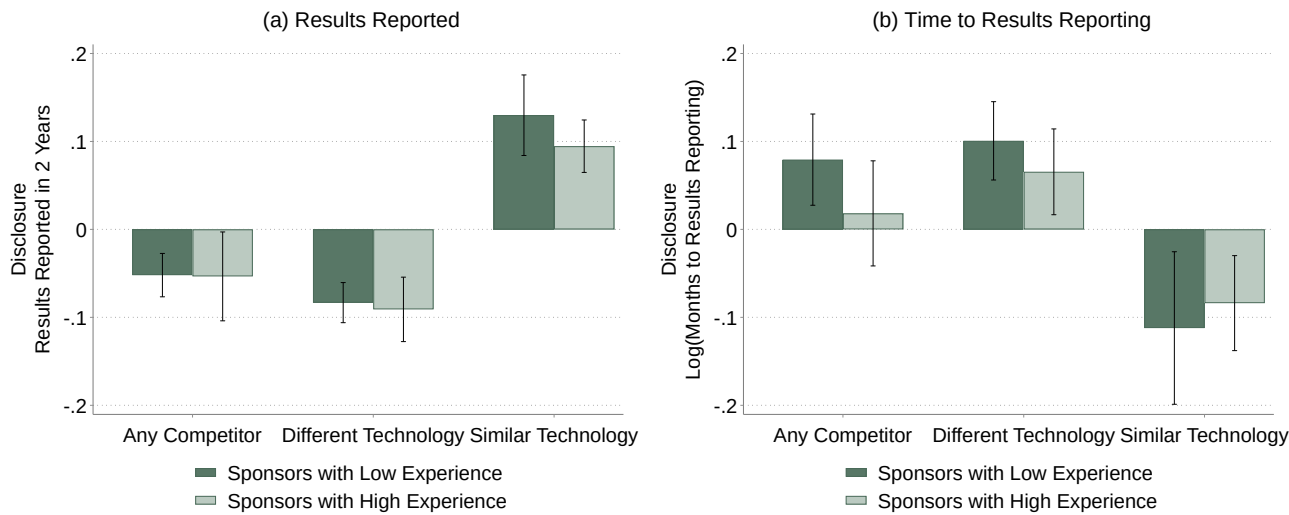
As a complement to our previous analyses, we also examine the types of firms that are most responsive to competitor priority review drug approvals; in particular, we examine the role of a firm’s drug development experience in shaping its strategic response. A large literature has highlighted the importance of firm research experience in shaping pharmaceutical firms’ decisions (see, e.g., Guedj and Scharfstein 2004, Chan et al. 2007, Arora et al. 2009, Hermosilla 2021, Krieger et al. 2022b). If firms make disclosure decisions in order to maintain and improve their competitive position, the likelihood that they would change their disclosure decisions in response to a competitor’s priority review drug approval may vary by their drug development experience and size. For example, smaller firms with little or no experience in successfully developing drugs in a specific product market may be more exposed to the impact of a competitor’s drug approval in that same market relative to larger firms with a larger portfolio of successfully developed drugs in that market. Anecdotal evidence suggests that smaller, less experienced firms may view a new successful competitor as a greater threat to its survival and therefore, may be more cautious about changing their disclosure decisions.<sup>40</sup>

Thus, we investigate whether a firm’s experience affects how it responds to the approval of a competitor’s drug in the same disease.<sup>41</sup> In particular, we categorize clinical trials into two mutually exclusive

groups: those conducted by smaller, less experienced firms and those conducted by larger, more experienced firms as proxied by trial sponsors with below or at versus above the median levels of clinical trials initiated in the prior five years in the same disease (MeSH code). Figure 4 presents coefficients from regressions that are fully saturated with controls (for detailed estimates, see Table E3 in the Online Appendix).<sup>42</sup>

The first set of bars in panel (a) of Figure 4 indicates that the impact of any competitor drug approval does not lead to noticeably different effects on the probability of reporting across different firm types. However, the first set of bars in panel (b) of Figure 4 indicates that smaller, less experienced firms are more likely to delay the reporting of trial results following competitor drug approvals. A deeper analysis of different types of competitor drug approvals reinforces these findings; except when deciding to reduce disclosure after a competitor approval based on different technology (the second set of bars in panel (a) of Figure 4), smaller, less experienced firms are more inclined than larger, more experienced firms to shift their disclosure strategies in response to competitor drug approvals.<sup>43</sup> The differences between the estimates are not statistically significant, likely because of several factors; the small sample size for each group and the constrained resources of smaller firms may limit their ability to fully adjust their disclosures. Nevertheless, these findings provide suggestive evidence that a firm’s experience may play a

Figure 4. (Color online) Impact of Drug Approvals: Heterogeneity by Sponsoring Firm Experience



Notes. This figure explores how the change on ClinicalTrials.gov results reporting following a competitor priority review drug approval varies by the type of competitor approval and firm drug development experience. A trial is classified as having a smaller, less experienced (larger, more experienced) sponsoring firm if it has at/below (above) median prior drug development experience in the focal MeSH code. The effects of the three types of competitor drug approvals on small, less experienced and larger, more experienced firms in the same MeSH code are examined. In panel (a), the outcome is a binary indicator for whether trial results are reported on the site within two years of trial completion. In panel (b), for reported trials, the outcome is the log of the time to results reporting. Shown are 95% confidence intervals. The specification is based on trial-MeSH observations, and the sample consists of private-sector trials initiated in/after 2007 and completed by 2019. OLS regressions include year fixed effects; MeSH code fixed effects; controls for trial phase, duration, and the number of participants enrolled; and disease market size. For more detailed variable descriptions, see Section 3.

role in determining how it uses strategic disclosure in response to a competitor's drug approval.

#### 6.4. Alternative Mechanisms

The previous evidence suggests that firms engage in strategic disclosure in response to competitor priority review drug approvals. Before proceeding to the final set of heterogeneity analyses, we wish to make one remark; the conceptual framework described in Section 2 suggests that firms disclose product quality information to maintain and improve their competitive position. In particular, competitor priority review drug approvals shift firms' competitive expectations, which increase their concerns about dampening consumer interest and lead to a reduction in disclosure. This contrasts with an alternative framework where changes in disclosure are primarily because of changes in product quality. Under this alternative framework, such competitor drug approvals may cause firms to raise their investment threshold (i.e., reduce investment in the number of lower-quality projects and invest primarily in higher-quality products) to minimize direct competition via prices (see, e.g., Shaked and Sutton 1982, Milgrom 2008, Gaynor 2011, Matsa 2011, Bloom et al. 2015). Put differently, such competitor approvals may reduce firm incentives to initiate investment in products that are unlikely to succeed, such as lower-quality projects. Early models suggest that firms primarily disclose results for clinical trials that exceed a given quality threshold (Viscusi 1978, Grossman and Hart 1980, Jovanovic 1982). This indicates that competitor priority review drug approvals shift firms' competitive expectations, raising their concerns about dampening consumer interest, and this could lead to an overall *increase* in the rate of disclosure among conducted trials.<sup>44</sup> Under this alternative mechanism, in response to competitor priority review drug approvals, product quality disclosure is just a proxy for project quality. In this section, we present three reasons for why this is unlikely.

First, our empirical findings contradict the expected changes in disclosure that would result from competitor drug approval-driven shifts in project quality. Our previous findings suggest that, on average, competitor priority review drug approvals lead to shifts in firms' competitive expectations. Under the alternative framework, such approvals would cause firms to increase their investment threshold (i.e., reduce investment in the number of lower-quality projects and invest primarily in higher-quality products), lowering the overall level of trials conducted. We expect an increase in both the average quality and the rate of disclosure among conducted trials. However, our empirical findings, which show an overall *decrease* in the rate of disclosure, directly contradict these expected outcomes. This provides suggestive evidence that in response to competitor drug approvals, changes in product quality

disclosure are not just a proxy for changes in project quality.

Second, consistent with the idea that competitor priority review drug approvals do not meaningfully shift the quality of rival firms' projects, Table E4 in the Online Appendix shows that there is no statistically significant impact of such approvals on firm project quality<sup>45</sup> as measured by the likelihood that a clinical trial successfully advances to the next stage of development.<sup>46</sup>

Third, our main findings—that competitor drug approvals lead to a decline in subsequent disclosure—are robust even when we conduct a disease-level analysis that holds the level of clinical trial investment fixed. Table E6 in the Online Appendix shows that following a drug approval in an MeSH code, the share of trials disclosing results declines. Note importantly that timing discrepancies also lower the likelihood of this alternative mechanism; changes in firms' R&D and quality investments likely occur over the medium to long term, whereas our findings indicate that firms are able to adjust their disclosure decisions relatively quickly.

Taken together, this body of evidence suggests that a competitor drug approval-driven change in R&D investments and project quality is unlikely to be the primary channel through which competitor drug approvals shape disclosure. These results support the view that when faced with competitor drug approvals, nondisclosure is not simply a signal of low product quality (or bad news). Instead, firms may strategically shift the disclosure of information to maintain and improve their competitive position.

#### 6.5. Additional Heterogeneity

Online Appendix D presents additional extensions on the DiD results. Table D3 in the Online Appendix shows that the impact of a competitor priority review drug approval on firms' disclosure decisions is driven by private-sector trials rather than public-sector trials. In line with the findings in Table C2 in the Online Appendix, these results are consistent with the idea that private-sector (profit-oriented) institutions respond more to shifts in competitive pressures induced by competitor drug approvals.

Table D4 in the Online Appendix compares the effect of competitor drug approvals among phase III and phase II trials. The differences are not statistically significant. Nonetheless, the point estimates suggest weaker effects among phase II trials relative to phase III trials. These findings are consistent with the view that firms in later development stages and thus, closer in the drug development timeline to competitors with approved drugs may experience greater shifts in their competitive expectations.

To address concerns that competitor drug approvals may primarily shape firms' decisions for clinical trials generating noisier evidence, we analyze the impact of

competitor drug approvals among well-designed clinical trials. It is widely believed that the design of a clinical trial plays an important role in shaping the clarity and usefulness of the information generated, with better-designed trials often providing clearer evidence of the drug's safety and efficacy (Ioannidis 2005). For example, patient randomization, in which participants are assigned to treatment and control arms by chance, allows researchers to generate clinical trial outcomes that are less susceptible to bias from patient selection. Table D5 in the Online Appendix confirms that the effect of competitor drug approvals holds<sup>47</sup> among double-blind and randomized trials.<sup>48</sup>

## 7. Conclusion

Identifying the factors that shape firms' disclosure decisions and separately, understanding how firms sustain their performance in dynamic environments are of substantial interest to researchers, managers, and policy-makers. Bridging these two areas has been challenging for researchers because of the difficulty of finding appropriate settings for empirical investigation. This paper fills this gap by leveraging the unique features of the pharmaceutical industry to examine the causal impact of competitor priority review drug approvals on firms' product quality disclosure decisions. We show that, on average, competitor drug approvals hinder the likelihood and timeliness of product quality information disclosure. A competitor priority review drug approval lowers the likelihood that clinical trial results will be reported by 13%. Using detailed data that exploit variation in project failure and success, competitor drug approval type, and firm drug development experience, we provide several sources of evidence that together support the view that in response to competitor drug approvals, firms' disclosure decisions are motivated at least in part by strategic factors. Faced with a successful rival, firms may selectively withhold bad news and disclose good news.

Our analysis has several limitations and points to several opportunities for future research. For example, we focus primarily on how competitor drug approvals shape disclosure on the extensive margin (i.e., the decision of firms to disclose *any* key clinical trial results). Future work may be able to explore how competitor drug approvals shape disclosure decisions regarding the *amount* of information that firms choose to report. For example, firms may selectively collect and disclose information on a narrow set of critical clinical trial outcomes rather than the complete set of FDA-recommended clinical end points. In addition, although trial sponsors have strong incentives to register their clinical trials on ClinicalTrials.gov (recall the site registration requirements discussed in Section 2.1.2), there may still be incomplete clinical trial registration. To the extent that this incomplete

registration is correlated with our competitor drug approval measures, our estimates may not fully capture the shifts in firms' R&D decisions associated with competitor drug approvals.<sup>49</sup>

Further, although we have discussed clinical trial reporting across several platforms (e.g., ClinicalTrials.gov, in scientific journals, and at a major medical conference), disclosure can take place through other channels, such as press releases, financial filings, pharmaceutical representatives, and advertisements. What are the unique incentives associated with results reporting across these channels? In general, is the model of disclosure among firms different from disclosure among academic institutions? We hope that future work can incorporate reporting across alternative communication channels and shed light on these topics.

In addition, future work should explore how firms' strategic disclosure choices interact with their other key decisions made by themselves and their competitors. For example, within competitive, R&D-intensive industries, a firm's decisions of whether and when to patent are critical choices with significant implications for its future profits (see, e.g., Anton and Yao 1994, Gans et al. 2008, Mihm et al. 2015, Hegde and Luo 2018). Patented inventions must be novel and nonobvious, and as a result, firms may withhold clinical trial results because results that are revealed prior to patenting could be considered as prior art, potentially jeopardizing future patents that rely on these clinical data (Marandett and Savage 2021, Masur and Ouellette 2023). Understanding the relationship between the relative timing of firms' patenting and disclosure decisions and how they are influenced by competitor drug approvals is an important avenue for future research. In addition, although our main focus is on the impact of branded competitor drug approvals on firms' disclosure decisions, future studies should examine how these disclosure decisions are influenced by competitors' own disclosure decisions and the activity of generic competitors (Jin 2005).

More broadly, we hope that we have provided a starting point toward the push to provide a more complete picture of how competitive dynamics interact with disclosure. As we emphasize throughout the paper, a challenge of empirical work on disclosure is the challenge of characterizing products whose information is not disclosed. Our current approach, which involves using project failure and success as a proxy for project quality, constitutes a key step toward empirically documenting the relationship between competitive dynamics, disclosure, and project quality. Nonetheless, developing a more comprehensive measure of project quality offers another promising (and challenging) area for future work.

Although a full welfare analysis is beyond the scope of our study, our finding that competitor drug approvals



lead to a strategic decline in product quality disclosure has important managerial and regulatory implications. Additional analyses that incorporate additional measures of market opportunity (e.g., patient demand, pricing) and consider the optimal amount of results disclosure remain promising areas for future research.<sup>50</sup> Our results suggest that strategic disclosure is an effective lever by which firms maintain and improve their competitive position. These decisions are economically significant; a simple back-of-the-envelope calculation shows that if there was no effect of competitor drug approvals, all else being equal, there would have been 1,372 private-sector trials that would have publicly reported their results within two years following clinical trial completion. This translates into clinical trial results reporting from \$20.8 billion in investments in clinical trials.<sup>51</sup>

Indeed, firms can design disclosure strategies that maintain their competitive position while taking into account the environment's regulatory requirements. To illustrate, we compute a back-of-the-envelope calculation for an estimate of the fines associated with missing reporting on ClinicalTrials.gov; at \$10,000/day, the penalty for withholding results of a clinical trial for one year would be \$3.65 million. This yearly penalty is small in comparison with potential yearly drug revenues that might be negatively impacted by disclosure. The 2021 revenues for AbbVie's Humira were \$20.7 billion; for Merck's Keytruda, they were \$17.2 billion, and for Bristol Myers Squibb's Revlimid, they were \$12.8 billion.<sup>52</sup> Against this backdrop, managers should carefully consider their strategic disclosure choices and convey the strategic nature of these choices to key stakeholders (e.g., employees, investors).

In addition, considering their strategic disclosure decisions, R&D managers can draw insights from our results to improve their R&D practices. Given the substantial costs and uncertainties in the R&D process, the ability of R&D managers to effectively incorporate external knowledge is essential (Cohen and Levinthal 1990, Cockburn et al. 2000). In considering whether to initiate, terminate, or advance a research project, managers may look to competitors' clinical trial results. To the extent that R&D managers rely on external information generated from clinical trials, our findings demonstrate that it is essential to account for the strategic considerations that shape the set of available clinical evidence.<sup>53</sup>

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## Endnotes

<sup>1</sup> Most recently, concerns over clinical trial transparency have been salient in the race to develop safe and effective treatments for coronavirus disease 2019 (Covid-19; e.g., *STAT* ("Increase transparency at the FDA: We need sunlight to fight the pandemic"), the *New York Times* ("Vaccine makers keep safety details quiet, alarming scientists"), *Nature* ("COVID vaccine confidence requires radical transparency"), and the *New England Journal of Medicine* ("Evaluating and deploying Covid-19 vaccines—the importance of transparency, scientific integrity, and public trust")).

<sup>2</sup> Whereas high-level trial outcomes may be inferred from trial continuation and termination decisions as well as the selective dissemination of information through platforms, such as press releases, financial reports, and firm websites, our analysis is specifically focused on firms' strategic disclosure of detailed data on ClinicalTrials.gov. This includes information of substantial value for stakeholders, such as detailed trial results relating to key safety and efficacy characteristics of the drugs being clinically tested. In robustness checks, we also examine results reporting in publications and conferences.

<sup>3</sup> In this paper, when referring to "project quality," we are focusing on whether a drug demonstrates sufficient drug safety and efficacy in the treatment or prevention of a disease, which are key Food and Drug Administration (FDA) criteria for approving new therapies (Dabrowska and Thaul 2018). However, a drug's quality can encompass a variety of other factors as well, such as an improved dosage schedule (e.g., once a day versus three times a day), a more convenient delivery system (e.g., oral versus intravenous delivery), reliable and less expensive manufacturing and storage processes (e.g., stability that is not affected by temperature), and novelty (e.g., a newer, more effective mechanism of action that is difficult to duplicate).

<sup>4</sup> Postapproval, the FDA may also require manufacturers to conduct phase IV trials to assess safety.

<sup>5</sup> More specifically, under FDA regulations, all clinical trial sponsors must register all nonphase I interventional studies of drugs, biologics, and devices that have at least one U.S. site, are manufactured in



and exported from the United States, or are conducted under an FDA drug application. For additional details about ClinicalTrials.gov, see <https://ClinicalTrials.gov/ct2/about-site/background>.

<sup>6</sup> Other policies reinforce ClinicalTrials.gov registration requirements. For example, since 2005, the International Committee of Medical Journal Editors has required prospective clinical trial registration on ClinicalTrials.gov, or another major clinical trial registry, as a condition of publication.

<sup>7</sup> See the template that trial sponsors use to submit trial results to ClinicalTrials.gov: [https://prsinfo.clinicaltrials.gov/results\\_table\\_layout/ResultSimpleForms.html](https://prsinfo.clinicaltrials.gov/results_table_layout/ResultSimpleForms.html).

<sup>8</sup> Delays of up to two additional years are permitted in some cases, such as if the clinical trial sponsor is aiming to seek regulatory approval for an additional indication.

<sup>9</sup> Other well-known examples include GlaxoSmithKline, which was fined \$3 billion by the Department of Justice for concealing safety information associated with its drug Avandia (Wilson 2011), and Roche, which was a defendant in a \$1.5 billion False Claims Act suit alleging that the firm concealed safety and efficacy information related to its drug Tamiflu (Dyer 2020).

<sup>10</sup> Firms are also incentivized to seek priority review as it can potentially expedite the FDA approval process. Under existing regulations, the FDA has committed to approving priority drugs within a time frame that is four months faster compared with the review period for standard drugs. However, the FDA has committed to these goals for just 90% of NDAs designated as priority review, and the period can be extended if the FDA, during their review, requires additional information, which can lead to additional rounds of review (U.S. Government Accountability Office 2020132).

<sup>11</sup> In a complementary analysis, we show that the days surrounding the approval of firms' priority review drugs are associated with positive market reactions, suggesting that novel and valuable information is released to the market. This is described further in footnote 31.

<sup>12</sup> Theoretically, a competitor's successful drug approval could reduce the number of available substitutes (thus, lessening competitive pressure) if other firms subsequently cease investment in clinical trials. However, our data indicate that this effect is unlikely to be sufficiently strong. Section 6.4 shows that priority review approvals have a limited impact on rivals' R&D activities.

<sup>13</sup> The approval of a competitor's priority review drug may also impact rivals by changing regulatory expectations, such as setting the newly approved drug as the standard of care in future clinical trials. However, two factors suggest that these regulatory changes are unlikely to be a first-order concern in this context. First, the analysis in Section 6.4 indicates that priority review drug approvals have limited effects on rivals' subsequent R&D efforts. Second, the FDA's history of approving multiple priority review drugs for the same indication implies that there is unlikely to be a limit to the number of drugs that can be approved for any given disease market.

<sup>14</sup> Although the drug development process can lead to first-mover advantages to firms that secure early approval (Lieberman and Montgomery 1988a), this setting differs from traditional patent races in that they are not winner takes all (Reinganum 1989). Typically, multiple drugs are approved to treat the same disease.

<sup>15</sup> This is based on the authors' calculations using Drugs@FDA and the Compilation of CDER New Molecular Entity Drug and New Biologic Approvals dataset.

<sup>16</sup> Highlighting the value of competitor information, following Merck's safety-related withdrawal of its drug Vioxx, Pfizer halted existing trials on Celebrex, its competitor to Vioxx, and redirected efforts toward assessing the drug's safety—activities that Pfizer could have avoided or initiated sooner had Merck disclosed Vioxx's safety risks earlier (Solomon et al. 2005, Vonkeman et al. 2006).

<sup>17</sup> Using MeSH codes to classify disease markets follows the methodology of the recent innovation literature (Azoulay et al. 2010, 2021; Koning et al. 2020, Myers 2020). Nevertheless, the main results are robust to using alternative trial-disease categorizations (for more details, see Section 5.3 and Online Appendix C).

<sup>18</sup> Higher participant enrollment numbers may increase marginal disclosure costs because of the additional costs of collecting and preparing data for each extra participant. Firms that rely on contract research organizations may avoid the need for substantial investment in their own reporting infrastructure, potentially leading to lower marginal reporting costs.

<sup>19</sup> This measure follows the literature (e.g., Budish et al. 2015) that uses the number of patients to account for market size. Future work should incorporate more direct measures of market opportunity by incorporating drug prices and expected patient demand.

<sup>20</sup> For simplicity, we categorize phase II/III trials—adaptive trials that combine phases II and III into a single trial—as phase III in our summary statistics and progression analyses.

<sup>21</sup> Drugs that do not advance from phase II to phase III or do not win FDA approval are considered failed drugs. The algorithm used to construct phase-specific clinical trial advancement measures required us to rely on the set of clinical trials that were mapped to standardized Cortellis drug and firm names. This resulted in a final "trial advancement data set" of 97,438 private-sector trial-MeSH observations.

<sup>22</sup> The objective response rate is commonly measured using Response Evaluation Criteria in Solid Tumors criteria. For more details, see <http://recist.eortc.org/>.

<sup>23</sup> These findings are consistent with recent laboratory evidence (Jin et al. 2021).

<sup>24</sup> In Online Appendix C, we show that our results are robust using the number of approved competitor drugs and other alternative measures of competitor activity.

<sup>25</sup> Figure B4 in the Online Appendix shows that clinical trials with high and low levels of competitor trials appear ex ante different on several observable factors; clinical trials in markets with more competitor trials are associated with trial sponsors with less R&D experience and more participants, and they are more likely to be phase III.

<sup>26</sup> Although common scientific and market opportunity shocks are unlikely (Krieger 2021), we include year fixed effects in  $X_{it}$ .

<sup>27</sup> Logit models yield very similar results, but linear probability models allow us to include more controls.

<sup>28</sup> In examining changes in the competitive landscape, we focus on the effects of drug approvals as other major changes (e.g., drug withdrawals) are less common.

<sup>29</sup> The control group consists of trial-MeSH observations that are not yet in diseases with a competitor drug approval.

<sup>30</sup> The Prescription Drug User Fee Act of 1992 requires that the drugs with priority review status must be approved within six months of submission of an NDA. However, most firms do not announce when they have submitted NDAs.

<sup>31</sup> Similar to Aryal et al. (2023), we conduct a stock market event study to explore the movements in a firm's market value around the days that a given firm's priority review drug is approved. Cumulative abnormal returns (CARs) measure the effects of unexpected changes in firm value (MacKinlay 1997). Our untubulated results show significantly positive CAR (2.5%) over a window of  $[-1, 1]$  days of a firm's priority review drug approval. Similar studies that conduct stock analyses around the timing of FDA announcements include Miller (2017) and Garfinkel and Hammoudeh (2024).

<sup>32</sup> Trials that are initiated prior to the competitor drug approval and concluded after it may also respond to the timing of the approval. Our main results are robust to the inclusion of ongoing trials

evaluating “contemporary” drugs (i.e., those that were previously initiated within one year prior to the competitor drug approval).

<sup>33</sup> The timings of these three events—drug marketing application submission, priority review designation, and drug approval—are all associated with considerable public ex ante uncertainty regarding their exact timing. Under existing regulations, following receipt of an NDA, the FDA conducts an initial 60-day review to determine the drug’s eligibility for priority review. As described in footnote 10, after the initial review, the FDA aims to approve priority drugs four months faster than the standard review timeline. However, the FDA has committed to this goal for just 90% of drug marketing applications. Additionally, the FDA can extend the review timeline if additional material is determined to be needed from the sponsor (U.S. Government Accountability Office 2020). Importantly, public information about each of these events differs substantially. Aryal et al. (2023) show that the public announcement of a drug’s approval—typically, a drug’s most profitable development point and an event that is always publicly disclosed—triggers an economically significant increase in a firm’s market value. In contrast, NDA submissions are not typically announced prior to approval and do not correspond to significant increases in the market value of the drug’s manufacturer.

<sup>34</sup> This qualifies proposed treatments for expedited review per the FDA’s Expedited Programs for Serious Diseases (Department of Health and Human Services 2014). This includes drugs participating in the following FDA programs: Priority Review, Fast Track, Accelerated Approval, and Breakthrough Therapy Designation.

<sup>35</sup> Table E1 in the Online Appendix shows that our results are robust to controlling for drug novelty, which may also impact firms’ advancement decisions (Krieger et al. 2022a).

<sup>36</sup> It is also possible that results from such trials are less conclusive, leading firms to be more careful about disclosing such results to public markets or competitors.

<sup>37</sup> We thank an anonymous reviewer for highlighting this point.

<sup>38</sup> These results are consistent with recent work by Garfinkel and Hammoudeh (Forthcoming), which shows that the response of pharmaceutical firms to their competitors’ outcomes varies based on the level of market competition.

<sup>39</sup> For example, in response to favorable clinical results for one of Novo Nordisk’s GLP-1 drugs, the head of Eli Lilly’s diabetes division noted, “The results are great for the anti-obesity medication class” (Pagliarulo 2023).

<sup>40</sup> Interviews with pharmaceutical industry experts conducted for this paper support the view that smaller, less experienced firms are much more risk averse, and this feature plays an important role in shaping their disclosure decisions. Such risk aversion may be driven by having less capital and the need to consider the effects of disclosure on fundraising and the possibility of eventual acquisition by a larger, more experienced firm.

<sup>41</sup> In addition to risk, small and large firms may also differ in the level of resources that they can allocate for disclosing trial results and the public pressure that they face to make such disclosures. Our empirical strategy examines the impact of competitor priority review drug approvals on firm disclosure net of these factors.

<sup>42</sup> Here, because regressions split the sample by at/below versus above median levels of sponsoring firm drug development experience, we do not directly include controls for sponsor experience.

<sup>43</sup> One concern might be that smaller, less experienced firms may not disclose if they cease operating after finding unfavorable trial results. However, this scenario would cut against the results documented in the third set of bars (corresponding to the impact of competitor approval with similar technologies) in both panels of Figure 4. This discrepancy suggests that these firm-exit effects are not sufficiently strong.

<sup>44</sup> This disclosure threshold can be a function of a variety of factors, including the products’ current sales and the risk associated with disclosure (e.g., the likelihood that disclosing negative product quality information will trigger a product recall).

<sup>45</sup> These results contrast with those of Garfinkel and Hammoudeh (2024), who find that in competitive settings, an increase in competition as measured by the FDA’s designation of a rival’s drug as a “breakthrough therapy” is associated with an overall decline in the likelihood that a clinical trial in the same product market advances to the next stage. Our analysis differs from that of Garfinkel and Hammoudeh (2024) in several key ways. First, they focus on drugs with breakthrough therapy designations, which are more select designations than priority review status (see Department of Health and Human Services 2014 for more details). For example, among drugs initially approved on and after 2012 (the year in which the FDA’s Breakthrough Therapy Designation program began), 60% were granted priority review, whereas 22% were designated as breakthrough therapies. Second, we focus on a change in competitor activity in the form of the *approval* of a rival drug with priority review status (not the preapproval designation of the drug as a breakthrough therapy). Finally, we examine trial advances for both phase II and phase III trials. Garfinkel and Hammoudeh (2024) focus on phase II trials.

<sup>46</sup> These results are robust to excluding trials testing drugs with similar technologies as those of the approved drugs, reducing concerns that heterogeneous effects among such trials may drive the overall results. See Table E5 in the Online Appendix.

<sup>47</sup> Our baseline results remain largely unchanged after controlling for trial design (see Table D7 in the Online Appendix). Note importantly that firms’ trial design decisions are endogenous, and competitor approvals may also shift firms’ trial design decisions to shift the noisiness of the information signals provided to competitors, with randomized trials providing clearer, more reliable signals. This interpretation is corroborated by Table D6 in the Online Appendix. That said, the extent to which firms shift their trial design decisions is also shaped by regulatory guidelines and their incentive to learn about their product’s underlying safety and efficacy.

<sup>48</sup> A caveat to these results is that trial design might not fully characterize the robustness of evidence produced. For example, although randomized and double-blind trials are often considered the most rigorous, in certain diseases and settings, the FDA may advise using historical controls or unblinded methods when testing new drugs. In such cases, regulators may deem these methods sufficient for assessing a drug’s safety and efficacy (e.g., just 51% of cancer drug trials used as the basis for FDA approval were randomized (Ladanie et al. 2020)). Additionally, nonrandomized and/or unblinded trials might unexpectedly result in more comprehensive data collection and be more likely to reach enrollment targets as they often require fewer participants.

<sup>49</sup> Current data limitations prevent us from characterizing extensive margin disclosures at scale and from identifying clinical trials that are not registered. Accordingly, these data limitations should be taken into account when interpreting the implications of our findings.

<sup>50</sup> There is a broad literature in economics, finance, and accounting that studies optimal disclosure. See Healy and Palepu (2001) for a review.

<sup>51</sup> We calculate these estimates using the number of private-sector trials that were “treated” by the competitor priority review drug approvals. There were 21,296 trial-MeSH observations tested in an MeSH code where a competitor priority review drug was ever approved. Table 3 shows that a competitor drug approval lowers the likelihood that a trial reports its results within two years following clinical trial completion by 13%. This suggests that 2,768 ( $\approx 21,296 \times 0.13$ ) trial-MeSH observations will not disclose results.

To convert this to the trial level, we note that trials in this sample are typically associated with two MeSH codes. Converting trial-MeSH observations to the trial level gives 1,384 trials. To estimate the dollar value associated with these 1,384 trials, we take the average of the reported phase II and phase III trial costs (\$15 million) from Martin et al. (2017), which results in an estimated \$20.8 billion.

<sup>52</sup> A complete analysis that incorporates the full set of potential regulatory fines and the probability of receiving a fine remains an important area for future research.

<sup>53</sup> “Missing” disclosures can increase the likelihood of project failures and wasteful spending among competitors, increasing R&D costs (notably, R&D failures are estimated to cost \$140 billion each year (Mullard 2023)).

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